Tweedle Formula

$$\boxed{\underset{\text{ruined}}{Y} = \underset{\text{clean}}{X} + \sigma \underset{\text{noise}}{Z}}$$

(정확한) 평균, 분산

$$\mathbb{E}[X \mid Y=y] = y + \sigma^2 \nabla_y \log p_Y(y)$$

$$\mathrm{Var}[X \mid Y=y] = \sigma^2 \left(I + \sigma^2 \nabla_y^2 \log p_Y(y) \right)$$

Note. 열역 흐르

$$\mathbb{E}[X \mid Y=y] = y + \underbrace{\sigma^2}_{\text{불확실}} \underbrace{\nabla_y \log p_Y(y)}_{\text{"가기가 가요"}}$$

---

① $p_{Y|X}(y|x) = \phi_\sigma(y-x)$

가우시안

② $\nabla p_Y(y) = \int p_X(x) \left(\nabla_y \phi_\sigma(y-x) \right) dx$

미분

$$= \int p_X(x) \left(-\frac{1}{\sigma^2} (y-x) \phi_\sigma(y-x) \right) dx$$

$\because$ $\phi_s = C \cdot \exp\left(-\frac{\|u\|^2}{2\sigma^2}\right)$

$$\nabla \phi_s = \underbrace{C \cdot \exp(\cdots)}_{= \phi_s} \underbrace{\nabla_u \left(-\frac{\|u\|^2}{2\sigma^2}\right)}_{-\frac{u}{\sigma^2}}$$

③ $$= -\frac{1}{\sigma^2} \left[y\, p_Y(y) - p_Y(y) \underbrace{\int x\, p_{X|Y}(x|y)\, dx}_{= \mathbb{E}_{X|Y}[X \mid Y=y]} \right]$$

$\because$ $p_X(x)\, p_{Y|X}(y|x) = p_{X|Y}(x|y)\, p_Y(y)$

④ $$\nabla \log p_Y(y) = -\frac{1}{\sigma^2} \left[y - \mathbb{E}_{X|Y}[X \mid Y=y] \right]$$

$\because$ $\nabla \log p_Y(y) = \frac{\nabla p_Y(y)}{p_Y(y)}$

$Y = X + \sigma Z$ when $Z \sim N(0, I)$

then if $\varepsilon \approx 0$

$p_{X|Y}(x|y)$ is Gaussian

---

① $$p_{X|Y}(X|Y) = \frac{\overset{\text{가우시안}}{p_{Y|X}(Y|X)} \cdot \overset{\text{추정}}{p_X(X)}}{\underbrace{\cancel{p_Y(Y)}}_{\text{const}}}$$

② $$p_{Y|X}(Y|X) = \underbrace{\cancel{\frac{1}{(2\pi\sigma^2)^{d/2}}}}_{\text{const}} \cdot \exp\left(-\frac{\|y-x\|^2}{2\sigma^2}\right)$$

---

③ $$p_X(x) \approx p_X(y) + \underbrace{\langle \nabla p_X(y), x-y \rangle} + O(\underbrace{\cancel{\|x-Y\|^2}}_{\text{small}})$$

$$= p_X(y) \langle \nabla \log p_X(y), x-y \rangle$$

$$= \cancel{p_X(y)} \left(1 + \langle \nabla \log p_X(y), x-y \rangle\right)$$

$$\approx \underbrace{\cancel{p_X(y)}}_{\text{const}} \cdot \exp\left(\overset{\text{추정}}{\langle \nabla \log p_X(y), x-y \rangle}\right) \qquad (1+x \approx e^x)$$

---

③ $$p_{X|Y}(X|Y) \propto \exp\left(\overset{\text{가우시안}}{-\frac{\|y-x\|^2}{2\sigma^2}} + \overset{\text{추정}}{\langle \nabla \log p_X(y), x-y \rangle}\right)$$

$$= (\cdots) = -\frac{1}{2\sigma^2} \left\| \begin{matrix} (x-y) \\ -\sigma^2 \nabla \log p_X(y) \end{matrix} \right\|^2 + \text{Const}$$

$$\approx N\left(y + \sigma^2 \nabla \log p_X(y),\ \sigma^2 I\right)$$

DDPM.

(forward) $x_t | x_{t-1} = \sqrt{\alpha_t}\, x_{t-1} + \sqrt{1-\alpha_t}\, z_t \sim N(0, I)$

$\approx N(\sqrt{\alpha_t}\, x_{t-1}, (1-\alpha_t) I)$

Note $\alpha_t + \beta_t = 1$

$0 < \beta_t < 1$

$$x_2 = \sqrt{\alpha_2}\, x_1 + \sqrt{1-\alpha_2}\, z_2$$

$$= \sqrt{\alpha_2}\sqrt{\alpha_1}\, x_0 + \sqrt{\alpha_2}\sqrt{1-\alpha_1}\, z_1 + \sqrt{1-\alpha_2}\, z_2$$

$$= \sqrt{\alpha_2}\sqrt{\alpha_1}\, x_0 + \sqrt{1-\alpha_1\alpha_2}\, \tilde{z}$$

$\because$ $X_1 \sim N(0, \sigma_1^2 I)$

$X_2 \sim N(0, \sigma_2^2 I)$

then $X_1 + X_2$

$\sim N(0, (\sigma_1^2 + \sigma_2^2) I)$

---

hence, in general forward pass

$$x_t | x_0 \sim N(\sqrt{\underbrace{\alpha_1 \cdots \alpha_t}_{=\bar{\alpha}_t}}\, x_0, (1 - \underbrace{\alpha_1 \cdots \alpha_t}_{=\bar{\alpha}_t}) I)$$

이것은 Tweedie Formula에서

다루는 그 유리와 같다.